

Ultimate Multi-Modal RAG System

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Project Overview

The **Ultimate Multi-Modal RAG (Retrieval-Augmented Generation)** project is an advanced AI system designed to ingest, process, and retrieve information from diverse data sources. Unlike traditional RAG systems that only handle text, this platform supports **Text, Web URLs, PDFs, and Images**, providing a truly multi-modal experience. It leverages **Pinecone** as the vector database to ensure high-performance similarity searches and data management.

Key Features

1. Comprehensive Data Ingestion

The system features a robust ingestion pipeline categorized into three main modules:

- **Text Ingestion:** Allows users to input raw text of any length, attach custom metadata, and automatically generate chunks and embeddings for storage.
- **Web Ingestion:** Scrapes data directly from a provided URL. It crawls the website, extracts relevant content, and processes it into the vector database in real-time.
- **File & Document Ingestion:** Supports **Markdown and PDF** files. For PDFs, the system intelligently separates text from embedded images, ensuring both are indexed for retrieval.

2. Advanced Vision & OCR Capabilities

One of the standout features is its ability to process visual data:

- **Text-Based Images (OCR):** Extracts data from documents like invoices and receipts to be stored as searchable text.
- **Graphical Images:** Uses vision models to generate detailed descriptions of images.
- **Image Vector Storage:** Stores image vectors alongside text descriptions, allowing users to search for "similar products" or visual matches.

3. Multi-Modal Retrieval Process

The retrieval engine is designed for accuracy and versatility:

- **Semantic Querying:** Users can query the system in natural language to find specific information across all ingested documents and images.

- **Visual Search:** Supports "Image-to-Image" search. By uploading a file, the system identifies and retrieves the most similar items from the database.
- **Rich Metadata Outputs:** Every result includes detailed metadata, such as the parent file source, page numbers (for PDFs), and specific description tags.

Technical Stack

- **Backend:** Python FastAPI
- **Frontend:** Next.js
- **Vector Database:** Pinecone
- **AI Orchestration:** LangChain / LangGraph
- **Models:** Multi-modal LLMs for vision, OCR, and text embeddings
- **Automation:** Integrated workflows for chunking and embedding pipelines

Future Roadmap

To further enhance the precision of the **DTWP (Data-to-Web-Process)**, I plan to integrate:

- **Hybrid Search:** Combining keyword-based and semantic search.
- **Re-ranking Models:** Implementing a secondary scoring layer for the most relevant results.
- **HyDE & Self-RAG:** Utilizing Hypothetical Document Embeddings to improve retrieval in complex scenarios.

DEMO VIDEO:

<https://www.loom.com/share/d4d022dbb42c4bfab4a539f6f5fb2f97>

<https://www.loom.com/share/62820955bd6f4338b475b61a1a8a0cd2>

Screenshots

Ingestion

POST/api/v1/ingest/textIngest Text

POST/api/v1/ingest/webIngest Website Content

POST/api/v1/ingest/fileIngest File

Ingest various kind of files. Currently supported file list:

Text

Markdown

Image

Docx

PDF

CSV

Excel

JSON

Parameters

No parameters

Request bodyrequired

multipart/form-data

file * required

string(\$binary)

Choose File | No file chosen

source_name

Source identifier (e.g., FAQ, Web, Manual)

string | (string | null)

Default Source

Send empty value

collection_name

string | (string | null)

default-collection

Send empty value

author * required

string

AI team

image_content_type

If image, select text or graphic content type

TEXT

WEB

FILE

WEBSITE URL

<https://www.mgi.org/contact/faq>

COLLECTION NAME

FAQ

AUTHOR

AI team

EXECUTE

CLEAR

Web ingestion completed successfully.

CHUNKS

9

Chunk 6

e4bfc286bcf3c1b48a2db010a6d62ad961008884b105b8117aa565d06f0f19d5

WEB_CONTENT

Question: What is your Hotline / Call Center Number?

Answer: Thank you for reaching out. Our hotline number is: 09666-777055

Question: How to contact regarding Internship / Job / Employment?

Answer: Thank you for reaching out to us. Kindly visit our website to know about the career opportunities: <https://www.mgi.org/> or send an email with your CV in the following address: career@mgil.org

Question: How can I avail dealership?

▼ METADATA

source_name	Web
collection_name	FAQ
author	AI team
source_url	https://www.mgi.org/contact/faq
title	FAQ Meghna Group of Industries
type	web_content
chunk_index	6

Chunk 7

b0779b3c9cb9e1e52aa3dea18ddadd37dac7953ce17b0729978cd96a94aedf5f

WEB_CONTENT

TEXT

WEB

FILE

FILE UPLOAD

Drag & drop files here

OR

BROWSE FILES

Supported: TXT, MD, PDF, DOCKX, PNG, JPG, WEBP, CSV, JSON

SELECTED FILES

8-Fresh-Jhal-Ch...
302 KB

gura-moris.webp
50 KB

No.1-Full-Cream...
117 KB

spices11.jpg
30 KB

SOURCE NAME

Default Source

COLLECTION NAME

default-collection

AUTHOR

AI team

IMAGE CONTENT TYPE

graphic

IMAGE CONTEXT

spices

RUNNING...

CLEAR

8-FRESH-JHAL-CHANACHUR-22G.PNG

DONE

CHUNKS

1

Chunk 1

alab82f2233cd54ebf25c8af46c447fc43df5068e512f15444a3e9a62bdef0b3

IMAGE

IMAGE PREVIEW

View



The image depicts a snack food package from a brand named "Fresh." The bag features a vibrant red background with a prominent logo at the top center, displaying the brand name in a stylized font. The text "আলু চানাচুর" is clearly visible in bright yellow beneath the logo, indicating that the product is potato-based snack (alu chanar).

The packaging is designed to attract attention, utilizing bold colors and a glossy finish. There are also graphic elements representing the snack itself, showcasing varying shapes and textures of the potato snack.

The bottom section of the bag includes nutritional or promotional information in smaller text, likely in Bengali.

Key features of the packaging:

- Main colors: Red, yellow, and white

GURA-MORIS.WEBP

DONE

CHUNKS

1

Chunk 1

c86766cbdf0839742cefb9bce3005d705b688b6b2e7f24da3b8dcf2b4dc0cff

IMAGE

NO.1-FULL-CREAM-MILK-2KG-PACK.PNG

LOADING

Retrieval Test

CLEAR HISTORY

08:29 PM

mgi contact number

08:33 PM

Retrieved 5 chunks for "mgi contact number".

<https://www.mgi.org/contact/faq>

Question: What is your Hotline / Call Center Number? Answer: Thank you for reaching out. Our hotlin...

Answer: Please send us your complaint at email: info@mgi.org or inbox us to our official Facebook pa...

Sports * Media Centre * News & Events * Newsletter * Commercial Advertisements ...

Question: How can I avail dealership? Answer: Thank you for your interest. Kindly let us know of yo...

08:33 PM

Type your query...

ATTACH

IMAGE SEARCH

SEND

Retrieval Test

CLEAR HISTORY

turmeric powder

08:24 PM

Retrieved 3 chunks for "turmeric powder".

IMAGE PREVIEW

View



spices The image features a package of turmeric powder. The bag is predominantly yellow, reflecting ...

IMAGE PREVIEW

View



string The image features a package of cumin powder. Here are the details: - **Package

Type your query...

ATTACH

IMAGE SEARCH

SEND

Retrieval Test

CLEAR HISTORY

Using the image retrieved when first prompting design the bag has a predominantly red background,...

08:25 PM

find similar product



No.1-Full-Cream-Milk-2kg-pack.png 177 KB

08:29 PM

Retrieved 3 chunks for "find similar product".

IMAGE PREVIEW

View



image The image features a vibrant, colorful packaging design for a product labeled "No. 1." The bag...

IMAGE PREVIEW

View

Type your query...

ATTACH

IMAGE SEARCH

SEND